

Portfolio Risk & Performance Review

Executive Portfolio Risk Report | Historical Analysis, January 2016 - July 2026

Executive Summary

The portfolio generated an annualised return of 15.34% during the research period from January 2016 to July 2026. At 28.29%, AAPL produced the best asset-level return; however, its annualised volatility was the greatest at 28.92%. The key trade-off between return and durability was highlighted at the portfolio level with a Sharpe ratio of 1.22, a maximum drawdown of -26.9%, and around 1% 95% daily VaR.

Business Problem

An important question regarding portfolio risk is posed by the analysis: Which resources influenced performance? To what extent were those rewards associated with risk? Which holdings exhibited sufficiently distinct behaviour to justify diversification? And how bad might the historical downturn get? The goal is to transform a set of securities into a repeatable risk assessment that a non-technical stakeholder may examine via a Power BI dashboard.

Methodology

Pandas and NumPy were used in Python to process historical market values for eight instruments that were gathered via yfinance. For APL, JPM, JNJ, KO, XOM, SPY, GLD, and TLT, trading dates were aligned, daily percentage returns were generated, and portfolio and asset-level metrics were computed using Python. For beta, SPY was used as the industry standard. In order to connect the numerical analysis and stakeholder-facing graphics, the resulting analytical datasets were subsequently created for Power BI.

What do the numbers say?

Observation	Evidence
Strongest asset return	AAPL: 28.29% annualised
Highest asset volatility	AAPL: 28.92% annualised
Lowest asset volatility	TLT: 14.75% annualised
Portfolio Sharpe	1.22
Portfolio maximum drawdown	-26.9%
95% daily VaR	-1.09%

Key Findings

1. **Return leadership came with higher risk.** AAPL generated the highest annualised return at 28.29%, while its volatility was also the highest at 28.92%. This makes return ranking alone an incomplete decision rule.
2. **The portfolio delivered attractive historical risk-adjusted performance.** A Sharpe ratio of 1.22 indicates favourable excess return relative to portfolio volatility under the assumptions used.
3. **Diversification was meaningful but imperfect.** AAPL/SPY correlation was about 0.74 and JPM/SPY about 0.70, while JPM/TLT was about -0.31 and SPY/TLT about -0.16. The contrasting behaviour of TLT added a different return pattern to the equity-heavy group.
4. **Downside risk remains material.** The maximum historical drawdown was approximately -26.9%. This captures a peak-to-trough loss that volatility alone does not describe.
5. **Market sensitivity varied across assets.** AAPL beta was about 1.20 and JPM about 1.08, while TLT was about -0.13. The portfolio therefore contained assets with materially different exposure to broad equity-market movements.

Metric	Result	Interpretation
Annualised Return	15.34%	Historical Portfolio growth rate
Annualised Volatility	12.58%	Portfolio return variability
Sharpe Ratio	1.22	Risk-adjusted performance
Maximum Drawdown	-26.9%	Worst historical peak-to-trough decline
95% daily VaR	-1.09%	Approx. 1.1% daily loss threshold in this calculation

Recommendations

1. **Evaluate allocation alongside return.** Do not increase exposure solely because an asset ranks highly on historical return; review volatility, beta and drawdown contribution at the same time.
2. **Monitor downside metrics together.** Use maximum drawdown for historical stress experience and VaR for a statistical daily-loss threshold rather than treating either metric as a standalone risk limit.

Limitations

The analysis is historical and hypothetical, so results do not establish future performance. Beta depends on the SPY benchmark and the selected observation period.